Charalambos Louca

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I study nonlinear light-matter interactions at atomically thin scales. Through materials engineering, light-trapping cavities, and ultrafast spectroscopy, I explore how fundamental physical interactions can enable low-energy, high-contrast optical functionality with societal relevance. I strongly value collaborative research and enjoy sharing knowledge while building inclusive, supportive teams across disciplines and institutions.

Work Experience

Research Associate — University of Cambridge *Jan 2024 – Jul 2026*
Ultrafast vibrational/optical spectroscopy in plasmonic nanocavities
Responsible for ultrafast visible-range rig
Nonlinear plasmonics and optomechanics of quantum emitters under extreme confinement. Optical switching with self-assembled monolayers
Led the redesigne and rebuilt of NanoPhotonics Center ultrafast spectroscopy setup, during the Cavendish laboratory relocation. (NanoPhotonics Centre - Prof. J. J. Baumberg)

Research Associate — Politecnico di Milano *Feb 2023 – Jan 2024*
Ultrafast spectroscopy of strongly coupled 2D semiconductors with micron-scale resolution
Femtosecond polaritonic switching
Electrically tunable ultrafast dynamics of hybrid excitons. (Ultrafast Spectroscopy group - Prof. Giulio Cerullo)

Research Associate — University of Sheffield *Oct 2022 – Jan 2023*
Optical studies of twisted TMDC heterostructures and magnetic 2D materials in and out of the strong light–matter coupling regime. (2DSheffield - Prof. Alexander Tartakovskii)

Education

Ph.D. Physics — University of Sheffield *Oct 2018 – Jan 2023*
Thesis : *Interlayer excitons and polaritons in atomically thin van der Waals heterostructures.*
Advanced optical spectroscopy ; 2D fabrication ; knowledge exchange with industry on electrically contacted substrates for 2D semiconductors.

MSc. Nanomaterials and Materials Science — University of Sheffield *Sep 2017 – Sep 2018*
Dissertation on tribology and laser patterning of DLC with nanodiamond and graphite-like coatings.
Passed with *Distinction* ; *highest score* in cohort.

BSc. Physics — Durham University *Oct 2014 – Jun 2017*

Teaching / Outreach

Student Mentoring/Training *2021-Present*

Mentoring several MSc/PhD students across institutions to foster growth and independence and co-supervising their thesis.

Training several students and visitors on 2D semiconductor fabrication and optics.

Facilitated laboratory demonstrations (ellipsometry and interferometry) as part of practical laboratory modules.

2D Materials Summer School *Summer 2019*

Designed and taught/marked a course at Summer School on 2D materials (theory + hands-on) for visiting undergraduates from KSU (Saudi Arabia) ;maintained mentoring contacts beyond the school.

Nanophotonics Summer School *Summer 2024-2025*

Led design and delivery of academic sessions on ultrafast spectroscopy during Nanophotonics Summer school in Cambridge.

Physics at Work - University of Cambridge *Sep 2024*

Engaging secondary school students with interactive demonstrations to inspire interest in physics.

Carmel Teacher Training Internship *Summer 2016*

Taught Physics and Mathematics to high-school students ; science communication and classroom management. Marked examination papers.

Skills

Research and Teamwork

Skill-set acquired from Sheffield, Milan, Cambridge enables to create a unique niche of research direction in nanotechnology.

Expertise in 2D Solid state physics, microcavity and polariton physics, plasmonic nanophotonics and ultrafast spectroscopy.

Hold ongoing collaborations with groups at the University of Sheffield, University of Manchester, the University of Cambridge, Universidad Nacional Autonoma de Mexico, EPFL and Stanford.

Experience in inspiring general public and young generation through college participation and outreach events.

Industry Knowledge Exchange

Led a project on knowledge exchange with Helia Photonics and Ossila Ltd. Communicate knowledge and combine capabilities with industry to develop a product.

Practical Laboratory Skills

— **Advanced optical Spectroscopy**
Ultrafast spectroscopy, Nonlinear Optics, OPO/OPA alignment, micro-PL, Dark Field spectroscopy, K-Space Imaging / Spectroscopy.

— **Nanofabrication**
2D Material fabrication. Large experience in developing novel stacking methods

Organic Solution Preparation : Organic polymer solutions, spin coating.

Self-assembly methods : liquid-liquid and liquid air transfer for plasmonic nanoparticle, QD and Molecular monolayers.

— **Liquid Helium/Nitrogen Cryogenics**
Regular use of liquid helium flow and bath cryostats for optical studies, cooling down samples and 9T magnets
— **Full Experimental Automation**
Writing of scripts in Python / Matlab and LabView for fully autonomous experiment operation and data analysis for experiments with large number of datapoints.
— **Simulation and Modelling**
Transfer-Matrix and FDTD simulations to enable the design of optical micro/nano cavities for experiment optimisation and analysis.

Journal Services

Peer Review : Nature Communications, Advanced Materials, Nano Letters.

Topic Coordinator for Special Issue :
"Nonlinear Optical Processes in Emerging Nanomaterials", *Frontiers in Nanotechnology.*

Awards/ Grants

College Research Associate - Lucy Cavendish College, Cambridge
Competitive college fellowship selected out of other Cambridge postdocs (Sep 2024 - Jul 2026)

Laserlab Europe — Visiting Researcher Travel Grant (Politecnico di Milano) (2021)

Henry Royce Institute — Visiting PhD Researcher Access Grant (University of Manchester) (2019)

Best PhD Poster Award, Graphene 2020 Conference (2020)

Highest score in MSc cohort, University of Sheffield (2018)

Scholarship for Undergraduate Studies, Morphou Municipality (2014) ;

Publications

Co-First † / Corresponding ‖ Author

— **C. Louca**, A. Bykov, T. Mykytiuk, A. Crookes, S. Zhu, A. Demetriadou, J. J. Baumberg, "Self-similar nested nanoparticle-on-mirror structures enable discrete optical modes and enhanced light emission" (2026)
(*First Author : Under Review*)

— **C. Louca**, C. Trovatello, M. Celebrano and A. Genco, "Editorial : Nonlinear optical processes in emerging nanomaterials." *Frontiers in Nanotechnology* **8**, 1807665 (2026).
(*Published Editorial for Special Issue - First Author : Manuscript Writing*)

— E. Lapiore†, **C. Louca**† *et al.*, "Electrically tunable ultrafast dynamics and interactions of hybrid excitons in a 2D semiconductor bilayer"*Nature Communications* **16**, 10710 (2025).
(*Published - First Author : Measurements and manuscript writing*)

— A. Genco[†], **C. Louca**[†] *et al.*, "Femtosecond switching of strong light–matter interactions in microcavities with two-dimensional semiconductors." *Nature Communications* **16**, 6490 (2025). Selected for SDG 9 Collection
(*Published - First Author : Sample preparation, measurements and manuscript writing*)

— Y. Wang, **C. Louca**, S. Randerson, X. Hu, P. G. Zotev, O. P. Chaundler, P. Bouteyre *et al.*, "Exciton-polaritons in a monolayer semiconductor coupled to van der Waals dielectric nanoantennas on a metallic mirror." *arXiv preprint arXiv :2506.04979* (2025). (*Under Review - Nonlinearity measurements and analysis, Manuscript writing*)

— C. Cruciano, D. Rocco, A. Genco, A. Tognazzi, A. Locatelli, L. Carletti, A. Fedorov, C. Trovatello, G. D. Blasio, I.Bargigia, **C. Louca** *et al.*, "Shaping the emission directivity of single quantum dots in dielectric nanodisks exploiting Mie resonances." *ACS Nano*, **19**, 3, 3500-3509 (2025).
(*Published - K-space spectroscopy measurements*)

— **C. Louca**, F. Gucci, A. Genco, D. Ruiz–Tijerina, J. Zultak, C. Cruciano, R. V. Gorbachev, A. I. Tartakovskii, S. Dal Conte, G. Cerullo, "Homotrilayer $MoSe_2$ for increased lifetime and oscillator strength of hybridised interlayer excitons." In *Advances in Ultrafast Condensed Phase Physics IV*, p. PC1299205. *SPIE* (2024).
(*Conference Proceeding*)

— A. Catanzaro, A. Genco, **C. Louca**[‖], D. R. Tijerina, D. Gillard *et al.*, "Resonant band hybridization in alloyed transition-metal dichalcogenide heterobilayers." *Advanced Materials* **2309644** (2024).
(*Published - Corresponding Author : Sample preparation measurements and manuscript writing*)

— A. Genco, **C. Louca**, O. Dogadov, C. Trovatello, A. I. Tartakovskii, S. Dal Conte, G. Cerullo, "Ultrafast dynamics of 2D materials and their heterostructures." In *2023 IEEE Nanotechnology Materials and Devices Conference (NMDC)*, pp. 171–172. *IEEE* (2023).
(*Conference Proceeding*)

— P. G. Zotev, Y Wang, D.A Penares, T.S Millard, S. Randerson, X. Hu, L. Sortino, **C. Louca** et al "Van der Waals materials for applications in nanophotonics.", *Laser Photonics Rev*, 2200957, (2023)
(*Published - Strong coupling analysis*)

— **C. Louca**[†‖], A. Genco[†‖], S. Chiavazzo, T. P. Lyons, S. Randerson, C. Trovatello, P. Claronino,*et al.*, "Interspecies exciton interactions lead to enhanced nonlinearity of dipolar excitons and polaritons in MoS_2 homobilayers." *Nature Communications* **14**, 3818 (2023).
(*Published - First Author, Corresponding Author : Sample preparation, measurements and manuscript writing*)

— A. Genco, C. Trovatello, **C. Louca**, K. Watanabe, T. Taniguchi, A. I. Tartakovskii, G. Cerullo, S. Dal Conte, "Ultrafast exciton and trion dynamics in high-quality encapsulated MoS_2 monolayers." *physica status solidi (b)* **260**, 5, 2200376 (2023). (*Published*)

— **C. Louca**, "Interlayer excitons and polaritons in atomically thin van der Waals heterostructures." *Doctoral Dissertation*, University of Sheffield (2023).
(*PhD Thesis*)

— T. P. Lyons, D. J. Gillard, C. Leblanc, J. Puebla, D. D. Solnyshkov, L. Klompmaker, I. A. Akimov, **C. Louca** *et al.*, "Giant effective Zeeman splitting in a monolayer semiconductor realized by spin-selective strong light–matter coupling." *Nature Photonics* **16**, 632–636 (2022).
(*Published - Sample Preparation and measurments*)

Conferences

— **Oral**, IOP Photon 2026, September 2026 **(Invited Speaker)**
— **Oral**, UK Semiconductors, July 2023
— **Poster**, OECS 18, June 2023
— **Oral**, MRS Fall Meeting, November 2022
— **Oral**, Programme Grant Away Day, September 2022
— **Oral**, Graphene Week 2022, September 2022
— **Oral**, Photon 2022, September 2022
— **Oral**, UK Semiconductors, July 2022
— **Poster**, Graphene 2022, July 2022
— **Oral**, Graphene Flagship Meeting, March 2022
— **Poster**, Graphene2020 Virtual Conference, October 2020 **(Award)**
— **Poster**, Nanophotonics Winter School, January 2020
— **Poster**, UK Semiconductors, July 2019
— **Poster**, Lord Porter Laser Laboratory 1st Annual Symposium, June 2019

Languages

— Greek (Native)
— English (Fluent)